EVALUATING THE IMPACT OF IMMERSSIVE TECHNOLOGIES ON STEM EDUCATION IN SRI LANKA'S SYLLABUS

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DECLARATION

I declare that this is my own work, and this thesis does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Signature
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Signature of the Supervisor: Date:

Signature of the Co-Supervisor: Date:

Acknowledgements

I would like to express my sincere gratitude to everyone who supported me in completing this research project on Evaluating the Impact of Immersive Technologies on STEM Education in Sri Lanka's Syllabus.

First and foremost, my heartfelt thanks go to Mr. Aruna Ishara Gamage, my project supervisor, and Mr. Nushkan Nismi, my co-supervisor, for their continuous guidance, encouragement, and invaluable technical insights throughout this journey. Their mentorship was instrumental in shaping the direction of this research and ensuring the academic rigor and quality of this thesis.

I would also like to extend my sincere appreciation to the ICT teachers and Grade 10 students who participated in this study, whose honest feedback and active engagement were crucial in evaluating the usability and educational effectiveness of the developed Augmented Reality application.

Special thanks to the Department of Information Technology and the Faculty of Computing at the Sri Lanka Institute of Information Technology (SLIIT) for providing the resources, infrastructure, and academic support that made this research possible.

I am also grateful to my project teammates for their collaboration and shared commitment throughout the broader research initiative. Working alongside them made this experience both meaningful and rewarding.

Lastly, I am deeply grateful to my family and friends for their unwavering patience, encouragement, and support during this challenging yet fulfilling journey. Your belief in me gave me the strength to persevere and deliver my best work.

Abstract

Information and Communication Technology education in Sri Lanka, particularly for Grade 10 students, faces significant challenges in developing conceptual understanding due to the heavy reliance on static two-dimensional textbook diagrams to explain abstract concepts such as computer generations, computer networks, and logic gates. To address this issue, an offline-capable marker-based Augmented Reality (AR) mobile application was developed using Unity 3D, transforming static ICT textbook diagrams into interactive three-dimensional visualizations optimized for low-end Android devices with 2–3GB RAM. The application integrates directly with government-issued ICT textbooks, allowing students to scan existing diagrams to trigger interactive 3D models of selected illustrations. Each of the three lessons, computer generations, computer networks, and logic gates features an interactive explanation of the lesson content, followed by a quiz designed to reinforce learning, encourage active participation, and assess the student's level of understanding before and after engaging with the material. A quasi-experimental evaluation was conducted with Grade 10 students, measuring conceptual understanding, student engagement, and teacher adoption through pre- and post-intervention assessments. Key results indicated improvements in student comprehension of abstract ICT concepts, with notable differences observed between experimental and control groups. Teacher feedback further highlighted the application's potential for practical classroom integration within Sri Lanka's resource-constrained educational environments. These findings suggest that curriculum-aligned, device-optimized AR applications can meaningfully enhance STEM education outcomes when embedded within structured pedagogical frameworks.

Index Terms— Augmented Reality, STEM Education, ICT Curriculum, Mobile Application, Marker-Based Tracking, 3D Visualization, Sri Lanka, Educational Technology

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
APIs	Application Programming Interfaces
AR	Augmented Reality
ICT	Information and Communication Technology
R&D	Research and Development
STEM	Science, Technology, Engineering, and Mathematics
UI/UX	User Interface / User Experience
VR	Virtual Reality

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INTRODUCTION

Background and Context

The effective teaching of Information and Communication Technology (ICT) at the secondary level presents a unique pedagogical challenge: the concepts students must master—data communication, network design, binary logic, and hardware architecture—are fundamentally invisible. They cannot be held, observed directly, or manipulated in the way that physical science experiments allow. In Sri Lanka, where Grade 10 ICT is taught as a core subject under the national curriculum, this challenge is compounded by persistent resource constraints. Computer-laboratory facilities are often shared among multiple year groups, internet connectivity is unreliable in rural regions, and school budgets rarely extend to specialised pedagogical tools beyond the government-issued textbook.

The textbook's static, two-dimensional diagrams therefore become the primary—and often sole—visual resource available to both teacher and student. Research consistently shows that such representations are insufficient for developing the dynamic mental models required to understand processes such as packet routing, logic-gate cascades, or the operational differences between computer generations. The result is surface-level, examination-oriented learning that does not translate into genuine conceptual understanding or practical competence.

Over the past decade, Augmented Reality (AR) technology has matured from a laboratory novelty into a practically deployable pedagogical tool. Unlike Virtual Reality, which replaces the real world entirely, AR overlays interactive digital content onto the physical environment through a device camera. This characteristic makes it particularly well suited to the classroom: a student can point a smartphone at a printed textbook diagram and instantly see it transform into an animated 3D model. The cognitive pathway from two-dimensional symbol to three-dimensional dynamic system—a leap that traditionally requires significant imaginative effort and teacher scaffolding—is bridged directly by the AR experience.

Research Gap

Although a growing body of literature documents the effectiveness of AR in science and technology education, critical gaps remain when viewed through the lens of Sri Lankan secondary schooling. Four primary deficiencies characterise the existing research landscape:

- Curricular Alignment Gap: Existing AR applications for ICT education target vocational or university-level curricula (e.g., Taufik et al., 2023; Cowling & Birt, 2016) rather than the national secondary ICT syllabus.
- Technological Accessibility Gap: Published implementations assume mid-to-high-end devices and stable internet connectivity—conditions that are atypical in Sri Lankan government schools.
- Geographical and Contextual Gap: Batista et al.'s (2020) systematic review found minimal representation of South Asian educational contexts across 14 reviewed studies.
- Pedagogical Integration Gap: No reviewed application was designed to work directly with government-issued textbooks, instead requiring custom printed markers or 3D-printed physical components.

1.3 Research Problem

The central research problem addressed by this study is: How can a marker-based Augmented Reality mobile application be designed and deployed to improve Grade 10 students' conceptual understanding of abstract ICT concepts within Sri Lanka's resource-constrained secondary-school environment, while integrating seamlessly with the existing government-issued ICT textbook?

1.4 Research Objectives

The study pursued the following objectives:

1. To design and develop ARLearn ICT, a marker-based AR mobile application aligned with the Sri Lankan Grade 10 ICT curriculum that operates fully offline on low-end Android devices.
2. To evaluate the application's impact on student conceptual understanding using a quasi-experimental pre-test/post-test design.
3. To assess usability and student engagement through validated instruments (SUS; Likert-scale questionnaire).
4. To examine teacher adoption readiness and pedagogical integration challenges through semi-structured interviews.

5. To contribute a replicable, low-cost AR integration model applicable to other resource-constrained educational contexts globally.

1.5 Significance of the Study

This research makes several contributions to the field of educational technology. First, it provides empirical evidence of AR effectiveness within a South Asian secondary-school context, addressing a geographic gap identified by Batista et al. (2020). Second, the ARLearn ICT application demonstrates that curriculum-aligned, offline-capable AR education tools are technically and pedagogically feasible on hardware budgets accessible to developing-country schools. Third, the study's mixed-methods evaluation framework—combining quantitative learning-outcome measurement with qualitative teacher-perspective analysis—offers a methodologically rigorous template for future AR intervention research in similar contexts.

1.6 Dissertation Structure

The remainder of this dissertation is organised as follows. Chapter 2 presents a comprehensive review of relevant literature encompassing AR in education, STEM pedagogy, and mobile learning in developing contexts. Chapter 3 describes the system architecture and technical design of ARLearn ICT. Chapter 4 details the functional and non-functional requirements, and the technology stack employed. Chapter 5 outlines the research methodology and evaluation design. Chapter 6 presents and discusses results. Chapter 7 concludes with recommendations and future work.

CHAPTER 2: LITERATURE REVIEW

2.1 Immersive Technologies in Education

The term 'immersive technology' encompasses a spectrum from Augmented Reality—which overlays digital content on the real world—through Mixed Reality, which allows digital objects to interact with physical surfaces, to Virtual Reality, which replaces the real environment entirely (Milgram & Kishino, 1994). Each point on this reality–virtuality continuum presents distinct affordances and constraints for educational deployment. Batista et al.'s (2020) systematic review of 14 studies published between 2010 and 2018 found that 64.29% of reviewed applications employed VR and 35.71% AR, with no comprehensive MR application identified. Across these implementations, immersive technologies consistently produced positive learning outcomes: improved student motivation was documented in six studies, improved learning facility in nine, enhanced understanding in four, and greater interactivity in three.

A critical instructional finding from the review was that 79% of successful applications employed discovery-based learning strategies—environments in which students actively construct knowledge through exploration rather than passive information reception. This pedagogical orientation aligns with constructivist theory (Vygotsky, 1978; Piaget, 1952), suggesting that the effectiveness of immersive technology in education may be as much a function of instructional design as of the technology itself.

2.2 Marker-Based AR for ICT Education

Marker-based AR—in which the device camera recognises a pre-registered image target and overlays corresponding digital content—is particularly well suited to textbook integration because existing printed diagrams can serve as natural markers, eliminating the need for additional materials. Taufik et al. (2023) demonstrated this approach with a marker-based Android AR application for computer-network hardware education in Indonesian vocational schools. Their quasi-experimental study ($n = 72$) yielded a statistically significant improvement in learning outcomes: the experimental group achieved a mean post-test score of 88.44 versus 80.69 in the control group ($p < 0.05$), with a normalised gain of 77.11% categorised as 'effective'. ISO 25010 usability testing confirmed satisfaction rates exceeding 82% across all stakeholder groups.

The pedagogical design challenges inherent in AR-based learning were elucidated by Cowling and Birt (2016), who combined physical 3D-printed network components with mobile AR simulation in a university networking course. While usability scores were high—memorability (5.00/5) and navigability (4.33/5)—the study identified a critical design risk: students who bypassed the scaffolded two-stage learning process in favour of immediate unrestricted AR exploration experienced confusion and failed to achieve the intended understanding. This finding motivated the progressive-disclosure approach adopted in the present study's application design.

2.3 AR for Network Visualisation

Sato, Sakamoto, and Shimada (2015) developed a professional AR platform for wireless sensor network management that visualised invisible network processes—link topology, traffic flow, and device states—through smartphone camera overlays. Performance evaluations confirmed response times of 0.1–0.5 seconds for information retrieval, establishing the technical feasibility of real-time AR for complex network visualisation. Although designed for professional rather than educational use, the platform's conceptual contribution—that AR can render inherently invisible data-communication processes observable and manipulable—directly informs the interactive packet-routing simulations implemented in ARLearn ICT.

2.4 Mobile Learning in Resource-Constrained Contexts

The proliferation of low-cost smartphones in developing economies has prompted substantial research into mobile-assisted language learning, science education, and skills training (Traxler, 2018). Key success factors identified across this literature include offline functionality, low device-specification requirements, localised language support, and alignment with existing curricular materials (Ally, 2019). The present study operationalises each of these factors: ARLearn ICT functions without internet connectivity after initial installation, is optimised for devices with 2–3 GB RAM, provides labels in both Sinhala and English, and uses government-issued Grade 10 ICT textbook diagrams as its marker set.

2.5 Research Gaps Summary

Table 2.1 synthesises the gaps identified across the four primary source studies and positions the present research in relation to each.

Table 2.1 – Research Gaps Comparison Matrix

Study	Context	Technology	Gap Addressed	Limitation
Batista et al. [2]	Systematic Review (14 studies)	AR, VR, MR	Overview of CS education outcomes	No South Asian context
Taufik et al. [4]	Indonesian Vocational Schools	Marker-based AR (Android)	Network hardware visualization	Vocational only; custom markers
Sato et al. [1]	Professional WSN management	AR overlay (smartphone)	Real-time network visualization	High-end hardware required
Cowling & Birt [3]	University ICT networking	Mixed Reality + 3D print	Scaffolded MR learning	Requires 3D printer; HE only
This Study	**Sri Lankan Grade 10 ICT**	**Marker-based AR (Unity 3D)**	**Curriculum-aligned, offline, low-end devices**	—

As the matrix demonstrates, no existing study has simultaneously addressed curricular alignment with a national secondary ICT syllabus, technological accessibility on low-end devices, deployment within South Asian educational infrastructure, and pedagogical integration with existing government-issued textbooks. This study addresses all four gaps.

CHAPTER 3: SYSTEM DESIGN AND ARCHITECTURE

3.1 Overall System Description

ARLearn ICT follows a marker-based Augmented Reality architecture implemented in Unity 3D (LTS 2022.3) with AR Foundation 5.x providing the cross-platform AR runtime. The system is structured around four primary layers, each with discrete responsibilities that together deliver the end-to-end AR educational experience.

The design philosophy was shaped by three precedents from the reviewed literature. The marker-based approach mirrors Taufik et al.'s (2023) successfully validated methodology while extending it to a secondary-school context and government-issued textbooks. The real-time network visualisation techniques are informed by Sato et al.'s (2015) performance-optimisation patterns. The progressive-disclosure instructional design responds directly to Cowling and Birt's (2016) observation that unguided AR exploration undermines learning outcomes.

3.2 System Architecture

The architecture is decomposed into four layers as described below.

3.2.1 Input Layer

Students direct their device camera at specific diagrams within the government-issued Sinhala Grade 10 ICT textbook. The application maintains a pre-compiled image-target database of 24 diagrams spanning the four core topic areas: network topologies (bus, star, ring, mesh, hybrid), logic gates (AND, OR, NOT, NAND, NOR, XOR), computer generations (first through fifth), and input/output device illustrations. The camera feed is processed in real time at up to 30 frames per second on the target hardware profile.

3.2.2 Processing Layer

Once a frame is captured, the Vuforia Engine 10.x image-recognition subsystem compares it against the target database. On a successful match (confidence threshold ≥ 0.85), the system retrieves the associated Unity scene bundle and computes the homographic transformation matrix required to anchor 3D content to the detected marker plane. The 3D Rendering Engine—Unity's Universal Render Pipeline configured for mobile—instantiates the relevant prefab and applies the anchor transform, producing a spatially stable AR overlay. All scene bundles are preloaded at installation, ensuring offline operation.

3.2.3 Output Layer

Students interact with rendered 3D content through a touch-gesture interface. Single-tap selects a component for contextual information; pinch-to-zoom adjusts magnification; drag rotates the model. For simulation content—packet routing, logic-gate cascades—students use on-screen sliders and toggle switches to vary inputs and observe real-time output changes. Contextual explanation panels are available in both Sinhala and English.

3.2.4 Analytics Layer

With parental and student consent, an anonymised analytics module logs session duration, diagrams scanned, simulation interactions, and error events (e.g., marker-not-detected). Data are stored locally and uploaded to Firebase on next network availability, enabling aggregate analysis of engagement patterns for iterative application improvement.

3.3 Marker Recognition Workflow

The recognition pipeline operates as follows: (1) the camera frame is captured and converted to greyscale; (2) Vuforia's FAST-feature detector identifies candidate keypoints; (3) ORB descriptors are computed and matched against the pre-built target database using a Hamming-distance threshold; (4) RANSAC homography filtering removes outlier matches; (5) a confidence score is computed—frames below 0.85 trigger a 'Marker not detected, adjust camera angle' UI prompt; (6) accepted frames receive the anchor transform and the corresponding Unity scene is activated. Average pipeline latency on a Redmi 10C (Qualcomm Snapdragon 680, 4 GB RAM, Android 12) was measured at 1.3 seconds, comfortably within the 2-second non-functional requirement.